Amazon probed over fake reviews
Amazon and Google probed over efforts to stop fake reviews as it could mislead shoppers.
http://dgit.in/ Jul21-57

Uber drivers charged for rides
Some Uber drivers were being charged for giving rides due to software bug.
http://dgit.in/Jul21-58

```
docList = index.
get(keywords[i]);
            for (String s :
docList) {
                result.
add(s);
            }
        }
        System.out.
println(result);
    }
}
```

How does this code work?

This code will determine the number of files in the given directory and give a provision for a user to type a single line or multi line query. If the word or a sentence exist in any of the document it will display the names of the files where that particular words from query exist. It will create a separate index file named index1.txt which will have mapping of all the extracted words from the document collection and the document names in which they exist. It will also create another index file named index2.txt which will have mapping of all the extracted words from the document collection and the document names in which they exist but stopwords will be removed here. Make sure you create a stopwords.txt file and add all the stopwords and save it in the same directory.

PORTER STEMMING ALGORITHM

Stemming is a technique used to extract the base form of the words by removing affixes from them. It is just like cutting down the branches of a tree to its stems. For example, the stem of the words eating, eats, eaten is eat. Search engines use stemming for indexing the words. One of themost popularly used stemming algorithm is Porter stemming algorithm which makes use of these convention of word reduction

```
SSES → SS
IES → I
SS → SS
S →
```

If user searches for the queries like "caresses", using porter stemming it will become "caress" If user searches for the queries like "ponies", using porter stemming it will become "poni". If user searches for the queries like "caress", using porter stemming it will become "caress." If user searches for the queries like "cats", using porter stemming it will become "cat." So when porter stemming algorithm is applied during the index search what it will do is search for the word "caresses" and also "caress". So the main root word will also be searched even if you have written a plural word for the same. One disadvantage of porter stemming algorithm is that the words produced after Porter Stemming may not always be the real words or sensible words. But majority of times errors don't occur because this stemming algorithm is the best as compared to other stemming algorithms and is considered to have least error rate.

Java Program for Inverted Index and Porter Stemming Algorithm

```
//author – Sachin Deepak Ver-
lekar
public class PorterStemming {

    public HashMap<String,
ArrayList<String>> index =
new HashMap<>();
    public HashMap<String,
ArrayList<String>> stemIndex
= new HashMap<>();
    public File folder;
    ArrayList<File> files =
new ArrayList<>();

    public void readFiles()
throws Exception {
        folder = new
File("E:\\infodata\\Porter-
StemmingFile");
        System.out.
println("[*] Contents of
"+folder.getName());
                String[] dir =
folder.list();
                System.out.
println("-------------------
-------------------------");
                int counter=0;
                for(String x :
dir)
                {
                        coun-
ter++;
                        System.
out.print(x + "  ");

                }
                System.out.
println("");
                System.out.
println("-------------------
-------------------------");
                System.out.
println("[+] Number of con-
tents in the directory
:"+counter);
        PrintWriter writer1
= new PrintWriter("E:\\info-
data\\PorterStemmingFile\\in-
dex1.txt", "UTF-8");
        PrintWriter writer2
= new PrintWriter("E:\\info-
data\\PorterStemmingFile\\in-
dex2.txt", "UTF-8");
        for (File file :
folder.listFiles()) {
            if (!file.isDi-
rectory()) {
                files.
add(file);
                Scanner sc =
new Scanner(file);
                while (sc.
hasNext()) {
                    String
word = sc.next();
                    if (!in-
dex.containsKey(word)) {
                        in-
dex.put(word, new
ArrayList<String>());
                        in-
dex.get(word).add(file.get-
Name());
                    } else {
                        in-
dex.get(word).add(file.get-
Name());
                    }
                    if (!ste-
mIndex.containsKey(PorterStem
mingLogic(word))) {
```